

Provisional title

The emergence of the interesting. Birth of ideas, understanding and singularity of forms

F. Pachet, AI researcher (PhD, Hdr, former research director in the private sector, Sony then Spotify)

This theme and these insights are based on my experience as an AI researcher, characterized by a constructivist and empirical approach. I have spent 40 years building AI systems to generate music and texts, obsessively and sometimes successfully. I would now like to try to understand what I am doing.

Thesis project

This thesis project addresses a question that is both simple and elusive: what makes an idea, a work, a sentence, a melody, or an object seem to us *interesting*? This question raises several problems that philosophy has addressed in distinct frameworks: the birth of ideas, from analyses of access to the intelligible in Plato and Aristotle to reflections on invention, abduction in logic, or discovery in Peirce, Poincaré, or Hadamard; the experience of understanding, from the Aristotelian theory of knowledge to analyses of judgment and meaning in Kant, Husserl, or Wittgenstein; the suddenness of certain self-evident truths, from Cartesian intuition or the *intuitive knowledge* from Spinoza to the changes in aspect analyzed by Wittgenstein; finally the status of memory, attention and value, particularly in Bergson, Merleau-Ponty, Goodman or Deleuze.

The theoretical framework of this work rests on a compromise: it seeks to articulate a philosophy of the subject (inherited from Kant) with a naturalization of the transcendental (as proposed by Jean Petitot). Furthermore, we integrate the contributions of the theory of complex dynamical systems, emphasizing that the properties of emergent phenomena are not reducible to the properties of their constituent parts. Finally, this research examines universal properties, as well as those limited to a specific context, such as Western music.

We can begin with a series of very common experiences. Sometimes an idea suddenly appears, a previously vague thought crystallizes abruptly, a problem becomes comprehensible, a formula imposes itself with a kind of self-evidence. Sometimes, too, an object holds our attention disproportionately: a song, a modulation, a literary passage, an image, a line of reasoning fascinates us to the point of demanding repetition, re-listening, interpretation, and analysis. Yet this fascination cannot be reduced to mere novelty, beauty in the classical sense, subjective preference, or an objective property that can be isolated within the object. It seems to stem from a more complex relationship between a form, a memory, a host of expectations, and a capacity for recognition.

The project argues that these phenomena stem from the same question: the emergence of the interesting. The interesting is not understood here as a worldly qualifier, but as a possible philosophical category, designating the moment when something stands out for the mind as a form, in the strong sense of a salient and structured organization, a meaning this term can

have in the Aristotelian or phenomenological tradition as well as, more recently, in the work of Jean Petitot. The ambition of this thesis will thus be to establish a *philosophy of the interesting* by showing that the birth of ideas, understanding and fascination with certain forms are not three independent problems, but three aspects of the same phenomenon.

The guiding question can be formulated as follows: under what conditions does something appear interesting to a subject? This question leads to several others. Is the interesting element the object itself, the subject who receives it, or a relationship between the two—a difficulty already at the heart of Kant's analysis of the judgment of taste, which stems neither from an objective property of things nor from a simple private preference? Does a captivating song possess inherent properties that explain its power, according to a hypothesis close to the ambitions of *Hit Song Science* Or does this power depend essentially on memory, history, culture, and the subject's expectations? Can we conceive of what is interesting without reducing it to either raw objectivity or the subjectivism of taste?

The project posits that what is interesting designates a relationship of salience (of attention?) between a form and a spirit that is historically, emotionally, and culturally constituted. Something appears interesting when it fulfills several conditions simultaneously: a certain internal coherence, without which it remains confused; a tension, a surprise, without which it dissolves into the already seen; and finally, a power of reinvention, that is, the capacity to open up a space for repetition, variation, and possible thoughts. What is interesting would thus be that which combines intelligibility, singularity, and fecundity.

This hypothesis will be taken further. Certain particularly interesting objects present themselves as singularities of form: they seem to necessarily resolve a problem that was not fully formulated before their emergence. They do not simply answer a given question; by their very existence, they bring into being the question they answer. In this sense, the interesting object, as an invented necessity, can be compared to Valéry, when he describes invention as the production of a form that discovers its own constraints, or to Bachelard, when he shows that certain advances in thought reconfigure the very field of problems.

One of the research's challenges will be to clarify the relationship between object, memory, and subject in the experience of the interesting. A first temptation would be to locate the interesting within the object; a second to reduce everything to the subject. This project will seek to move beyond this opposition. It will hypothesize that the *interesting* is a relational property in the strongest sense: it depends on the subject's history, expectations, memory, culture, and perceptual abilities; but it also presupposes in the object an organization, an economy, a structure of tensions and resolutions. What is of interest is an encounter between a form and a memory capable of experiencing its singularity. On this point, the project may draw upon Bergson's analyses of memory and duration, Husserl's descriptions of retention and protention, or even Merleau-Ponty's concept of meaning as emergence within a field.

Music constitutes a privileged domain here. As a temporal object, it lays bare the relationships between memory, expectation, surprise, and recognition. Some songs fascinate because they seem to establish their own space of constraints, then resolve the expectations they have created in a way that appears obvious in retrospect but was impossible to anticipate. Particular attention will be paid to certain songs belonging to what I

propose to call, in a still exploratory manner, the regime *Jotney*. That is to say, forms where harmonic and melodic surprise is accompanied by a remarkable impression of local necessity. Music will thus function not as a mere particular case, but as a philosophical laboratory of the interesting.

Another important aspect will be the relationship between interest, rarity, and difficulty of production. It's not simply a matter of saying that what is interesting is what is rare in a statistical sense, but that there are forms where one senses that one could not move an element without altering the whole (self-stable). Added to this is sometimes an impression of surface simplicity that conceals a great difficulty in invention. This phenomenon, which can be linked to the notion of *difficulty* (developed in another context), can be reformulated philosophically: certain forms strike us because they seem self-evident, while suggesting the improbability of their genesis.

The method: conceptual analysis, study of authors, comparison of models of intelligibility, invention, and value. It will draw on cases from philosophy, literature, mathematics, and music. The project may also incorporate certain contemporary works in cognitive science and the psychology of *insight*, in perception theory or attention modeling, without, however, naturalizing the question. A previous project devoted to *interestingness*, developed within a computational framework, can play a conceptual background role here, provided that its intuitions are taken up and reformulated within a properly philosophical framework.

The provisional organization of the research could follow five phases: a phenomenology of the interesting; an analysis of understanding as the emergence of the intelligible; a reflection on the birth of ideas; a theory of the interesting singularity; and finally, a testing of these hypotheses in the case of music.

The originality of this thesis lies in its constitution of a philosophical object rarely treated as such: the interesting. Where philosophy has often privileged truth, beauty, goodness, meaning, or judgment, the aim here is to take seriously this intermediate and decisive category, which seems to govern a large part of our intellectual and aesthetic life, and even our everyday existence. The challenge is to propose a conception of thought that reduces it neither to the application of rules, nor to the simple reception of content, nor to logical reconstruction after the fact. The philosophy of the interesting would then be a renewed way of thinking together about invention, understanding, and value.

Initial bibliography

- Aristotle, *Metaphysical* (trans. Tricot), Vrin, 1991; *Seconds Analytics* (trans. Pellegrin), GF Flammarion, 2005 ; *Poetic* (trans. Dupont-Roc/Lallot), Seuil, 1980.
- Bachelard, G., *The formation of the scientific mind*, Vrin, 1999 (1st ed. 1938).
- Bergson, H., *Creative evolution*, PUF, 2013 (1st ed. 1907); *Thought and movement*, PUF, 2013 (1st ed. 1934).
- Berlin, D.E., *Studies in the New Experimental Aesthetics*, Hemisphere Publishing, 1974.
- Deleuze, G., *Difference and repetition*, PUF, 1968.

- Descartes, R., *Rules for the direction of the mind*, Vrin, 2002; *Metaphysical Meditations*, Vrin, 1992.
- Goodman, N., *Languages of Art: An Approach to a Theory of Symbols*, Hackett Publishing, 1976 (1st ed. 1968).
- Hadamard, J., *Essay on the psychology of invention in the mathematical field*, Gauthier-Villars, 1959 (1st ed. 1945).
- Husserl, E., *Experience and judgment*, PUF, 1970 ; *Formal logic and transcendental logic*, PUF, 1957.
- Kant, E., *Critique of Pure Reason*, PUF, 2001 ; *Critique of the faculty of judgment*, Vrin, 1993.
- Koestler, A., *The Act of Creation*, Arkana, 1989 (1st ed. 1964).
- Merleau-Ponty, M., *Phenomenology of perception*, Gallimard, 1945.
- Peirce, C.S., *Philosophical Writings of Peirce*, Dover Publications, 1955.
- Plato, *The Republic*, GF Flammarion, 2016 ; *Theaetetus*, GF Flammarion, 1995.
- Poincaré, H., *Science and Method* Flammarion, 1908.
- Schmidhuber, J., *Driven by Compression Progress: A Simple Principle Explains Essential Aspects of Subjective Beauty, Novelty, Surprise, Interestingness, Attention, Curiosity, Creativity, Art, Science, Music, Jokes*. In G. Pezzulo, M. V. Butz, O. Sigaud, G. Baldassarre (Eds.), *Anticipatory Behavior in Adaptive Learning Systems*, Springer, LNAI 5494, 2009.
- Spinoza, B., *Ethics*, Seuil, 1966.
- Valéry, P., *Notebooks*, Gallimard, 1973-1974 ; *As It Is*, Gallimard, 1943.
- Wittgenstein, L., *Philosophical research*, Gallimard, 2004 (1st ed. 1953).

Personal work related to the topic

Full list of publications: <https://www.francoispachet.fr/publications>

- Pachet, F. (2026) *Virtuosity within reach of poodles*, Hermann, forthcoming.
This text focuses on the different forms of virtuosity and their intended effects on listeners.
- Pachet, F. (2026) *On the impossibility of creating anything*, Matériologiques, forthcoming.
This essay lists all the scientific reasons opposing the very idea of creating something new.
- Pachet, F. Roy, P. (2026). *Markov Constraints and Controlled Sequence Generation*, in *Analysis Techniques for Mathematical Physics, Geometry, and Music*, Lecture Notes in Mathematics series. Edited by Nicholas D. Alikakos and Cédric Villani, to appear.
Synthesis of my previous work on style modeling and its combinatorial aspects.
- Pachet, F. Roy, P. Carré, B. (2021) *Assisted music creation with Flow Machines: towards new categories of new*. Chapter 18 of “*Handbook of Artificial Intelligence for Music*”, Eduardo Miranda ed., Springer.
I describe the main results of the Flow Machines project, and conclude with the idea that AI renews the categories of the new.
- Pachet, F. (2018) *History of an ear*, Buchet-Chastel.
This book (enhanced with Spotify QR codes) is an attempt to describe the ontogenesis of a musical ear. Alternative title: Proust Listens to the Beatles.

- *Pachet, F. (2012) Musical Virtuosity and Creativity. In McCormack, J. and d'Inverno, M, editor, Computers and Creativity, Springer*
I present virtuosity as an important feature of artistic productions, and describe a system (Virtuoso) that allows the generation of virtuoso phrases in bebop in real time.
- Pachet, F. (2008) former ERC project on the *interestingness* and related works on singularity, the difficulty of production and the relationship between creation and perception.
- Pachet, F. (2008) Machines for stepping outside oneself. In Bernard Stiegler, editor, The design of our existences in the age of upward innovation, pages 229-251, Mille et une nuits, Fayard.
In it, I describe Flow Machines, and the idea of building systems that promote Flow states according to Csíkszentmihályi
- Pachet, F. (2006) The keys to an interesting melody. The Birth of Art, pages 193-201, Tallandier.
This paper proposes (already) various approaches to modeling the interesting in melodies.